TOKAI Susano Divine eyes and feet. Precision rescue.

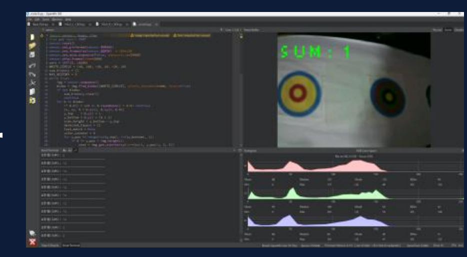
M64

HARDWARE SPECIFICATIONS

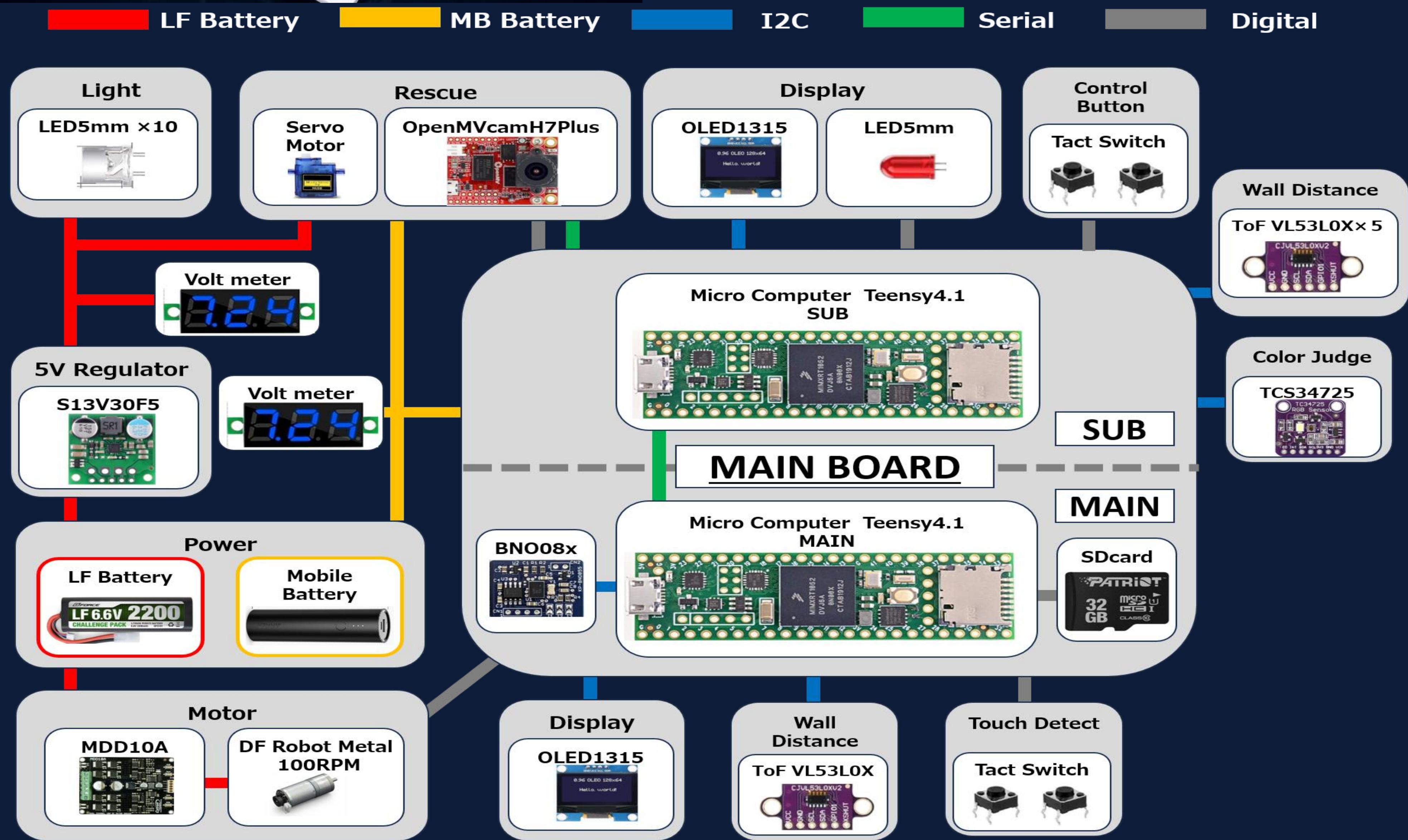
Dimensions: L210 x W198 x H207mm
Weight: 1,550g
Main Controller: Teensy 4.1 (816MHz) x2
Motor: RB-Dfr-759 x4 (6V DC 6.5kg-cm 100RPM)
Motor Driver: MDD10A x2
Vision System: OpenMV H7 Plus x2
Battery: • LF6.6V [2200mAh]
• Anker PowerCore 5V [5000mAh]
Sensor: • ToF Distance Sensor (VL53L0X x6)
• Color Sensor (TCS34725 x1)
• IMU(BNO08x x1)
Display: OLED SSD1315 x2
Servo Motor: SG90 x2 **LED:** x6(for Display)
Button: Tact Switch x3 **Touch Sensor:** Tact Switch x2

VICTIM DETECTION

Two OpenMV H7 Plus cameras detect victims on the left and right walls.
Letter victims Φ/Ψ/Ω are recognized by an image model trained with multiple fonts and rotations.
Recognition targets are detected as circular candidates, then the upper half is scanned vertically. LAB values are converted into color codes, the color-layer sum is calculated, and a 5-frame majority vote stabilizes the result. The result is sent to the main board by UART and used for rescue kit deployment.



Round Target → Vertical Scan → LAB Color Code → Layer Sum → 5-Frame Majority Vote → UART



NAVIGATION ALGORITHM

Frontier search, Backtracking & BFS

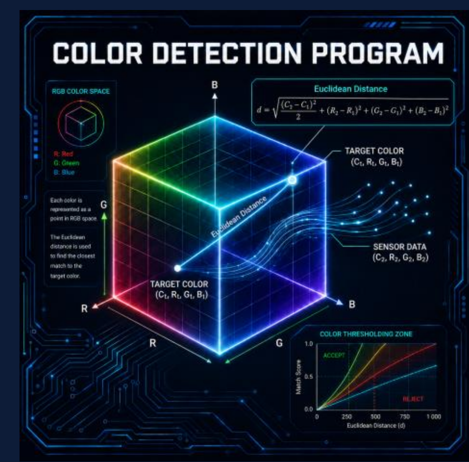
2	①	★	①
3	※	1	※
④	※	2	✓
✓	※	③	✓

★: Current position
※: Unexplored tile
1–4: BFS distance
✓: Explored
①: Selected path (left priority)

COLOR DETECTION

C, R, G, and B values are treated as a vector in color space, and a **weighted Euclidean distance** is calculated. (C weight: 1/2)

$$d = \sqrt{\frac{(C_2 - C_1)^2}{2} + (R_2 - R_1)^2 + (G_2 - G_1)^2 + (B_2 - B_1)^2}$$

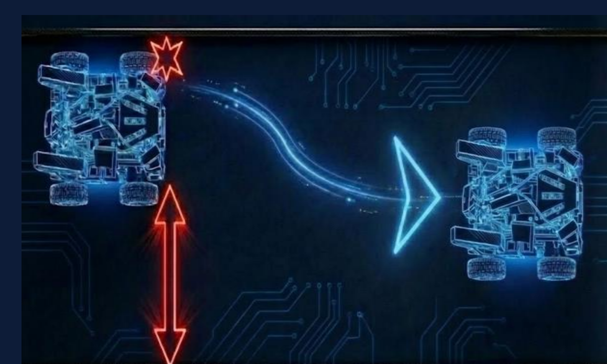


BACKUP SYSTEM

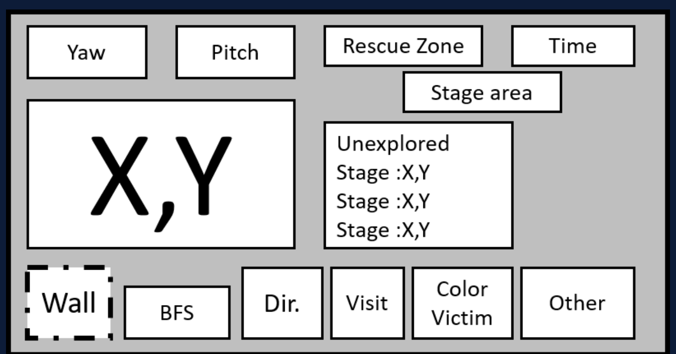
Category	Function
Localization	Red-zone exit localization
	SD-card map matching on restart
	Re-check uncertain wall detection
	Re-check ramp entrance/exit walls
Action	Timeout fallback action
	Periodic gyro correction & centering
	Color tile re-check
	Recovery mode when no route remains

WALL TRACE & GYRO TRACE

PD-controlled center driving
The robot keeps the center of the corridor using the distance difference between the left and right walls.
Early Detection
Angled front sensors detect the presence or absence of walls early.
Gyro Trace
When no wall is available, the robot follows its heading using the gyro sensor.



XY COORDINATE TRACKING



Exploration area: "+" active search "-" unexplored or explored
Return when both become "-"
Other: bump, ramp, stairs

MISSION COMPLETE ARCHITECTURE

Start
- Confirm saved data
  - Yes : Load map or start new
  - No : Mission Start
- Scan 4 walls & read color
- Move forward/ Color check at 7/13cm

Normal Zone Search
- Record red tiles; do not enter
- Blue tile: save position to SD
- Frontier search all Normal Zone areas
- Move to red tile by BFS

Red Zone Search
- Red tile: save to SD. No more saves.
- Frontier search all Red Zone areas
- Move to red tile by BFS or left-hand rule

Return
- Red tile: use SD to fix pos. & heading. Retry if unclear.
- Return to start by BFS, avoiding blue tiles
- At start: stop 10 sec and blink LED 10 times Mission Complete

RESCUE KIT MECHANISM

① Servo-controlled arms securely hold the rescue kits.
② The arms rotate inward and move to the loading position.
③ As the arms return, one kit is pushed out and dropped. Steps ①–③ are repeated.

Susano(素戔嗚): A hero of Japanese mythology who defeated Yamata-no-Orochi and saved a princess. We inherit that spirit and challenge rescue in the maze.
Yatagarasu (八咫烏): The three-legged crow said to have guided Emperor Jimmu to victory. It symbolizes guidance toward the correct path and our pledge for reliable, rapid rescue.
Kusanagi-no-tsurugi(草薙の剣): A legendary sword found in the tail of Yamata-no-Orochi. Like this sword, our technology cuts through difficulty and opens a path through the maze.